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Faculty of Computer Science and Engineering



Database Systems
Group 1 - CC02

Assignment 1

HOSPITAL DATABASE SYSTEM

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Section 1

Introduction

1.1 Context

This hospital database project aims to design a structured system to manage key entities within a hospital. The goal is to facilitate efficient management of different entities: storing, retrieving, and organizing large amounts of data while maintaining accuracy and integrity.

This database design must also reflect the relationships between various actors (e.g., employees and patients) and resources (e.g., departments, medication). By using Enhanced Entity Relationship (EER) model and transforming it into a relational schema, the database must maintain consistency and scalability.

By analyzing the database specifications, we aim to break down two primary elements of the (E)ER model: the distinct entities within the system and the relationships between them.



1.2 Entities and Relationships

1.2.1 Strong Entities

- Employees:
 - Doctors: professionals who diagnose, treat, and examine patients.
 - Nurses: assist doctors and care for patients.
- Speciality: an area of medical expertise, such as cardiology or neurology.
- Departments: divisions within the hospital, such as Rehab or HR.
- Patients:
 - Inpatients: admitted to the hospital for at least one night for treatment.
 - Outpatients: visit the hospital for medical care without staying overnight.
- Health Care Service:
 - Treatment: provided to patients to cure or manage illnesses.
 - Examination: performed to diagnose a patient's condition.
- Medication: a type of medications prescribed to treat patients.
- Prescription: specify medication and dosage, issued by doctors.
- Providers: supply medication to the hospital.
- Batch: contains medication supplied by provider.

1.2.2 Weak Entity

There are no weak entities in our database design.

1.2.3 Relationships

Relationships between entities are also crucial for the design. Firstly, we begin by identifying **binary relationships**:

No	Relationship	Description
1	works_for	An employee works for a department
2	is_specialized_in	An employee is specialized in some specialities.
3	manages	A dean, who is a doctor, manages a department.
4	care	Nurse takes care of inpatients
5	has	A treatment or examination has a medication (or prescription).
6	consists_of	A prescription consists of some types of medications with some quantity.
7	imported_from	A medication type is imported from batches.
8	supplies	A provider supply some batches of medication.

Table 1.1: All binary relationships in the hospital database



To represent the relationship between patients, doctors and the medication that the doctors provide, we introduce 2 **ternary relationships**:

No	Relationship	Description
1	treats	An inpatient can receive treatments from at least one doctor.
2	examines	An outpatients can have many examinations with their examining doctor.

Table 1.2: All ternary relationships in the hospital database.

To mitigate complex relationships that can make the database harder to manage, understand, and query effectively, the design avoids relationships that involve four or more entities. Thus, we focus entirely on binary and ternary relationships.



Section 2

Enhanced Entity Relationship Diagram

The introduction section enables us to develop an (E)ER diagram to capture key entities and relationships. It is the foundation of our database design.



2.1 Strong and weak entities

Entity name	Attribute	Description
Employee	Ecode Fname Lname Address Gender Dob Start_date Phone	Primary key: Employee's code Single-valued: First name Single-valued: Last name Single-valued: Address Single-valued: Gender Single-valued: Date of birth Single-valued: Start date Multi-valued: Phone numbers
Doctor (subclass of Employee)	-	-
Nurse (subclass of Employee)	-	-
Department	Dcode Dtitle	Primary key: Department's code Single-valued: Department's title
Patient	Pid Fname Lname Address Gender Dob Phone	Primary key: Patient ID Single-valued: First name Single-valued: Last name Single-valued: Address Single-valued: Gender Single-valued: Date of birth Single-valued: Phone number

Inpatient (subclass of Patient)	Date_of_disc	Single-valued: Date of discharge
	Date_of_adm	Single-valued: Date of admission
	Fee	Single-valued: Fee
	Room	Single-valued: Room
	Diagnosis	Single-valued: Diagnosis
Service	SID	Primary key: Service ID
Examination (subclass of Service)	Fee	Single-valued: Examination fee
	Diagnosis	Single-valued: Diagnosis
	Date	Single-valued: Examination date
	Next_exam_date	Single-valued: Next examination date
Treatment (subclass of Service)	Result	Single-valued: Treatment result
	Period_start	Single-valued: Start date
	Period_end	Single-valued: End date
	Recover_F	Single-valued: Recovery factor

2.2 Relationships

Notation:

- Partial: —
- Total: =

a. Binary relationships

- 1:1 relationship: `Manages`, `Has`.
- 1:N relationship: `Works_for`, `Care`, `Supplies`, `Is_specialized_in`.
- M:N relationship: `Consists_of`, `Imported_from`.



Manages

Dean, who is a doctor, manages a department.

Associated entities	Doctor, Department
Attributes	None
Cardinality	1:1 Each department has one dean. An employee must belong to a specific department and only a doctor can be a dean (managing a department). Therefore, a doctor can only manage one department.
Membership	Doctor: — Department: = Not every doctor is a dean. Every department must have a dean.

Has

A healthcare service (examination, treatment) that involves a medication prescription.

Associated entities	Service, Prescription
Attributes	None
Cardinality	1:1 Each treatment or examination has a prescription. A prescription belongs to a treatment or examination.
Membership	Service: — Prescription: = Whenever performing a healthcare service, the doctor(s) need not to give any medication to the patient. A prescription must belong to a treatment or an examination.



Works_for

Employee works for a department.

Associated entities	Employee, Department
Attributes	None
Cardinality	N:1 Each department can have many employees. An employee works for a specific department.
Membership	Employee: = Department: = An employee must work for a department. A department has at least one employee working for.

Care

Nurse takes care of inpatient.

Associated entities	Nurse, Inpatient
Attributes	None
Cardinality	1:N An inpatient is taken care of by a nurse. A nurse can take care of many patients at the same time.
Membership	Nurse: — Inpatient: = Each inpatient is taken care of by a nurse. A nurse can take care of many patients at the same time



Supplies

Provider supplies a batch of medication to the hospital.

Associated entities	Provider , Batch
Attributes	None
Cardinality	1:N A batch of medication belongs to the supply of one provider. A provider can supply many batches of medication to the hospital.
Membership	Provider : — Batch: = A provider may have not supplied medication to the hospital yet. A batch of medications must be provided by a provider.

Is_specialized_in

Employee is specialized in a speciality.

Associated entities	Employee , Speciality
Attributes	Degree 's year: The year that the employee is qualified for the speciality.
Cardinality	1:N A particular speciality degree belongs to one employee. An employee can be specialized in many specialties.
Membership	Employee : = Speciality: - It is assumed that either a nurse or a doctor must be specialized in at least one speciality. A speciality can be qualified by none working in the hospital.



Consists_of

A medical prescription consists of medication.

Associated entities	Prescription, Medication
Attributes	Quantity: amount of medication in the prescription.
Cardinality	M:N A medication can be in many prescriptions. A prescription can have many types of medications in it. This is because each treatment or examination can have medication(s), and a treatment/examination associated with a prescription.
Membership	Prescription : = Medication: - A medical prescription must have medication(s) in it. A medication needs not to be mentioned in any prescription.



Imported_from

A type of medication is imported from a batch of medication.

Associated entities	Medication, Batch
Attributes	Imported_price: The price at which the medication was imported in the batch. Imported_quantity: The amount of a type of medication that was imported from the medication batch. Exp_date: Expired date of a type of medication which is imported from the batch. Exp_flag: If medication from a batch is expired, the flag is 1 and is 0 otherwise.
Cardinality	M:N A medication can be in many prescriptions. A prescription can have many types of medications in it. This is because each treatment or examination can have medication(s), and a treatment/examination associated with a prescription.
Membership	Medication: = Batch: = Medication is provided by a provider (unit is batch). A batch must consist of medication(s).



b. Ternary relationships

There are 2 ternary relationships in the EERD: Treat and Examines.

Treat

Doctor(s) give treatment to an inpatient.

Associated entities	Inpatient, Treatment, Doctor
Attributes	None
Cardinality	1:N:N A doctor and a treatment has one inpatient. An inpatient can receive a treatment from many doctors. An inpatient and a doctor can have many treatments.
Membership	Inpatient: — Treatment: = Doctor: — An inpatient may not have any treatment yet when they are admitted to the hospital. A treatment always occurs when doctors treat inpatients. Sometimes, a doctor has no inpatient to treat.

Examines

A doctor examines an outpatient.

Associated entities	Outpatient, Doctor, Examination
Attributes	None
Cardinality	1:1:N A particular examination of one doctor corresponding to one outpatient. An examination of one outpatient is conducted by a doctor. The outpatients can have many examinations with their examining doctor.
Membership	Outpatient: — Examination: = Doctor: — An outpatient may not have any examination yet. A doctor may give no examination to any outpatient. An examination occurs when a doctor examines an outpatient.

2.3 Constraints

2.3.1 Constraints of specialization

Patient

- Inpatient and Outpatient are predicate-defined subclasses of Patient: we specify the condition of membership in the Inpatient and Outpatient subclass by the condition that their Pid must begin with “OP” and “IP” respectively.
- Inpatient and Outpatient subclass must be disjoint and total subsets of Patient: A patient must be classified as either a inpatient or an outpatient, but not both. Employee superclass:

Employee

Nurse and Doctor are user-defined subclasses of Employee.

- Nurse and Doctor subclass must be disjoint and total subsets of Patient: an employee must be classified as either a nurse or a doctor, but not both.
- Nurse and Doctor subclass must be disjoint and total subsets of Patient: an employee must be classified as either a nurse or a doctor, but not both.

Service

- Examination and Treatment are user-defined subclasses of Service.
- Examination and Treatment subclass must be disjoint and total subsets of Patient: In the context of the hospital here, a service that hospital offers to patients must only be either a Treatment or an Examination.

2.3.2 Semantic constraints

- Unique code for patients must follow the format “OP” or “IP” followed by 9 digits, specifically “OP” for outpatients and “IP” for inpatient.
- A doctor who manages a department must have a speciality related to that field and must have been awarded that speciality degree for more than 5 years.
- Medications out-of-date are automatically marked in the database.
- When a inpatient is recovered, his or her last treatment must be recorded and marked as “Recovered”.
- Discharge date of a patient can only be filled once their last treatment is confirmed as “Recovered”.
- All `End_date` values in Treatment must greater than or equal to their corresponding `Start_date` value for each record.



2.3.3 Domain constraints

- All the dates and dob must have a valid format “YYYY-MM-DD”.
- All the Degree’s year values of Speciality must have “YYYY” format.
- All the phone number must be in valid format “+84” followed by 9 digits (for the context that the hospital is in Vietnam).
- All of the Price values of the Medication entity, imported_price and imported_quantity of belongs_to relationship, quantity values of consists_of relationship and fee values of Inpatient entity must be greater than 0.

2.3.4 Constraint on NULLs

Entity type	Non-NULL Attribute (primary key excluded)
Employee	- Fname - Lname - Start_date - Dno
Department	- Dtitle - Mgr_code
Inpatient	- Date_of_asm - Nurse_code
Patient	- Fname - Lname - Phone
Medication	- Name
Belongs_to	- Imported_quantity - Imported_price - Exp_date
Batch	- Provider_num - Imported_date
Provider	- Name - Phone
Treatment	- Start_date
Examination	- Date



Section 3

EER-to-Relational Mapping

In this section, we map the (E)ERD to a relational schema, transforming the conceptual into a logical design.

3.1 Mapping of Strong Entity Types

In this step, each strong entity type in the EER model is mapped to a relation.

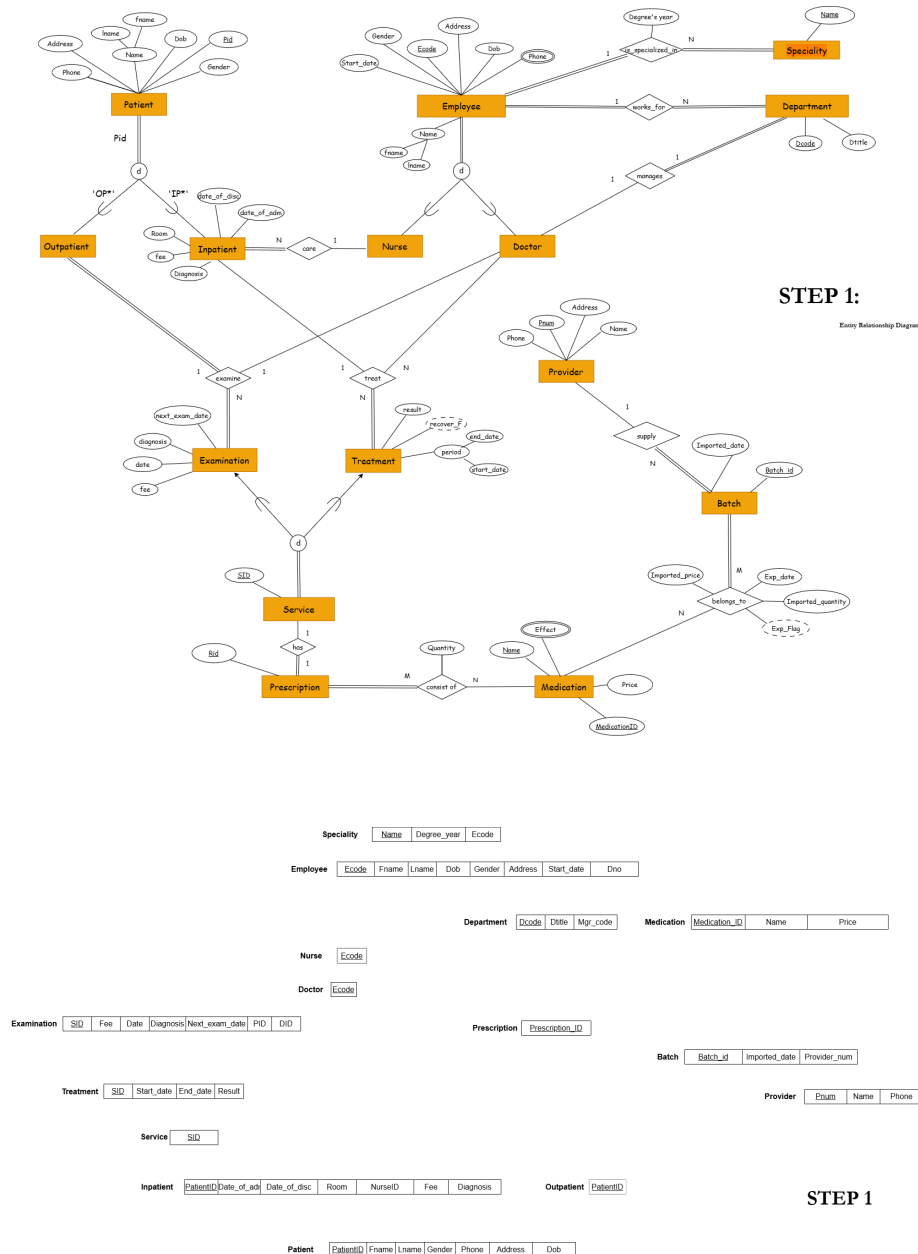


Figure 3.1: Mapping of Strong Entity Types

3.2 Mapping of Weak Entity Types

There are none of weak entity types in our database so we omit this step.

3.3 Mapping of Binary 1:1 Relationship Types

For binary 1:1 relationships, we can add the primary key of one entity as a foreign key of the other.

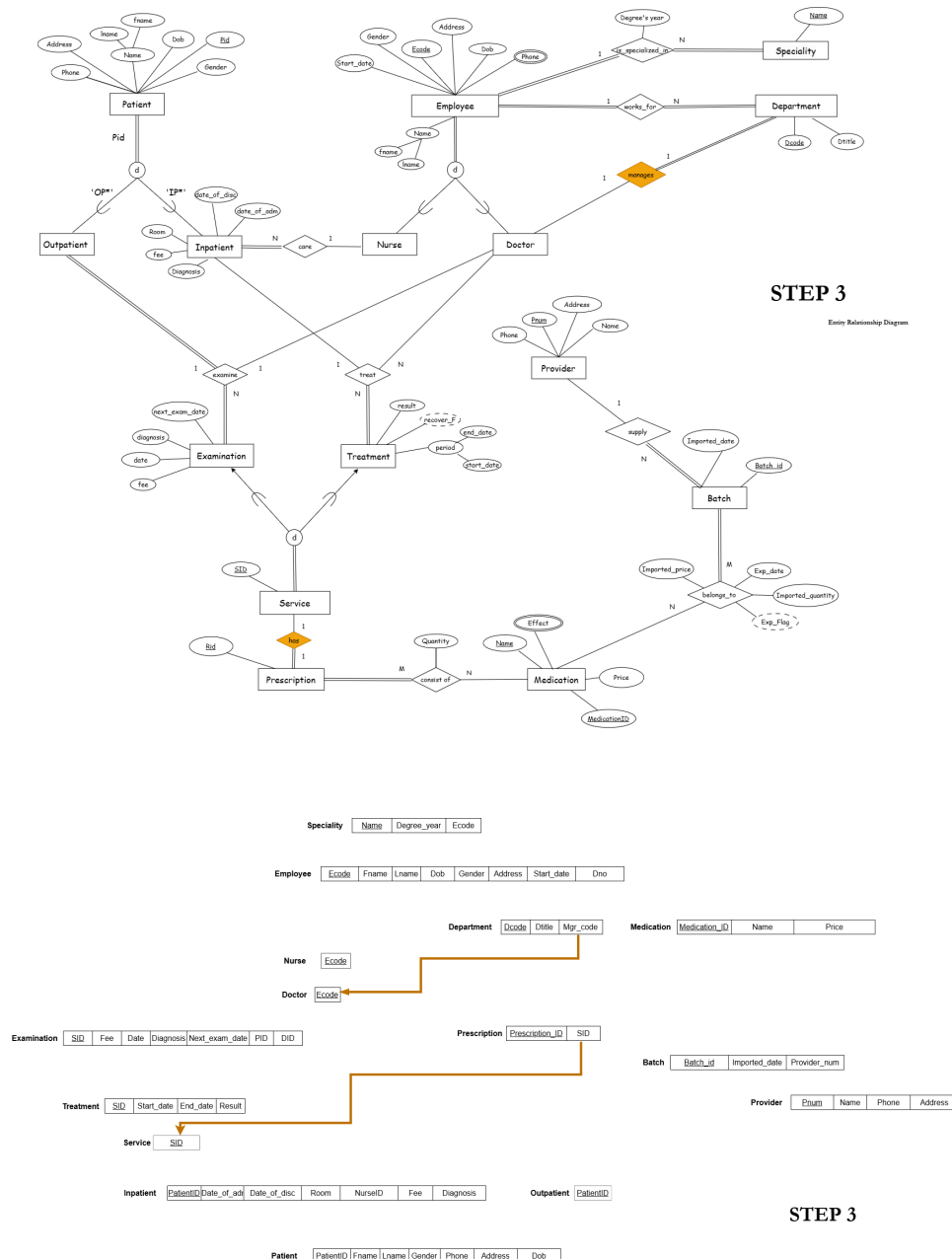


Figure 3.2: Mapping of Binary 1:1 Relationship Types

3.4 Mapping of Binary 1:N Relationship Types

In a 1:N binary relationship, we include the primary key of the "1" side entity as a foreign key in the relation representing the "N" side.

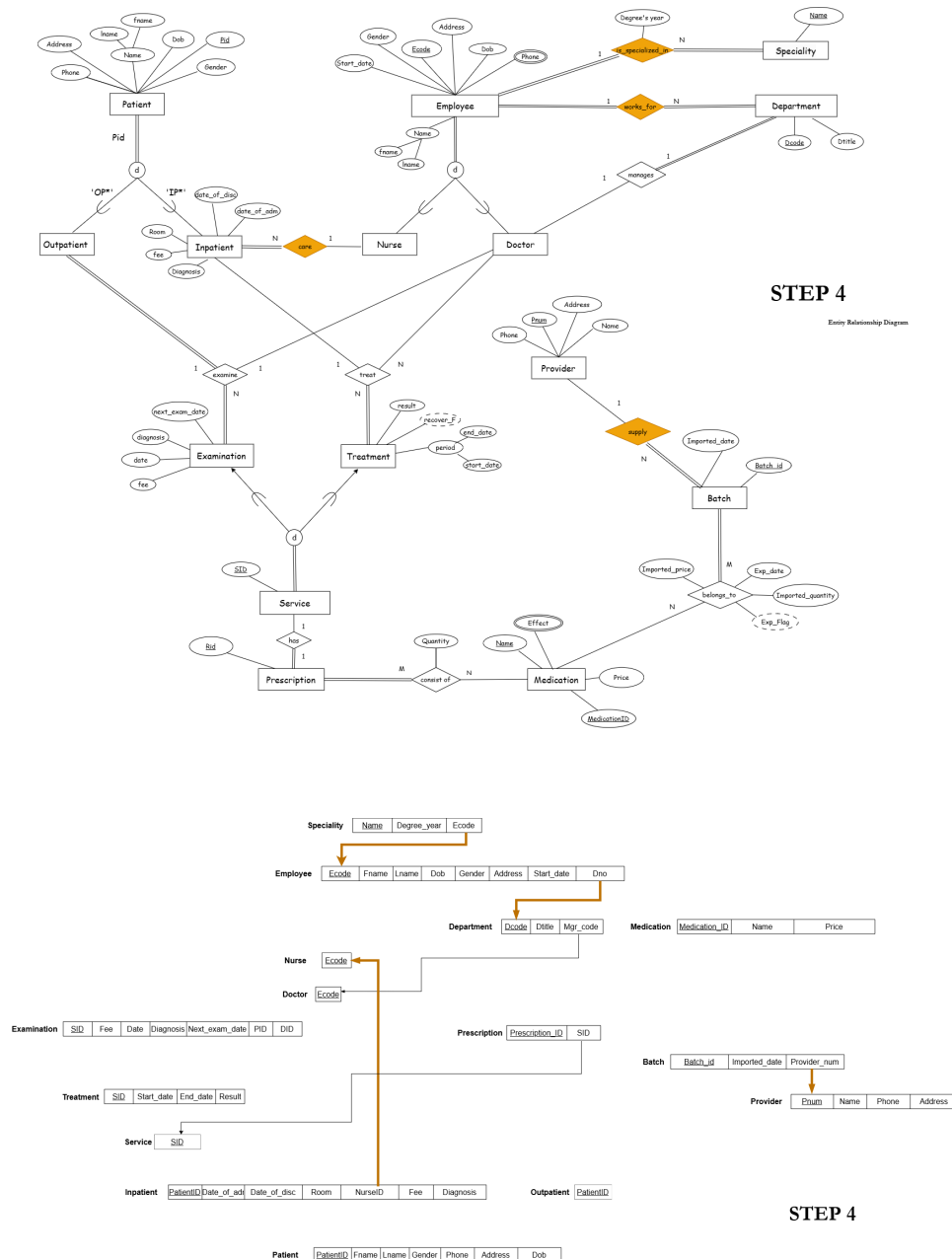


Figure 3.3: Mapping of Binary 1:N Relationship Types

3.5 Mapping of Binary M:N Relationship Types

For M:N relationships, we create a new relation (table) to represent the relationship. The primary keys of both participating entities become foreign keys in the new relation.

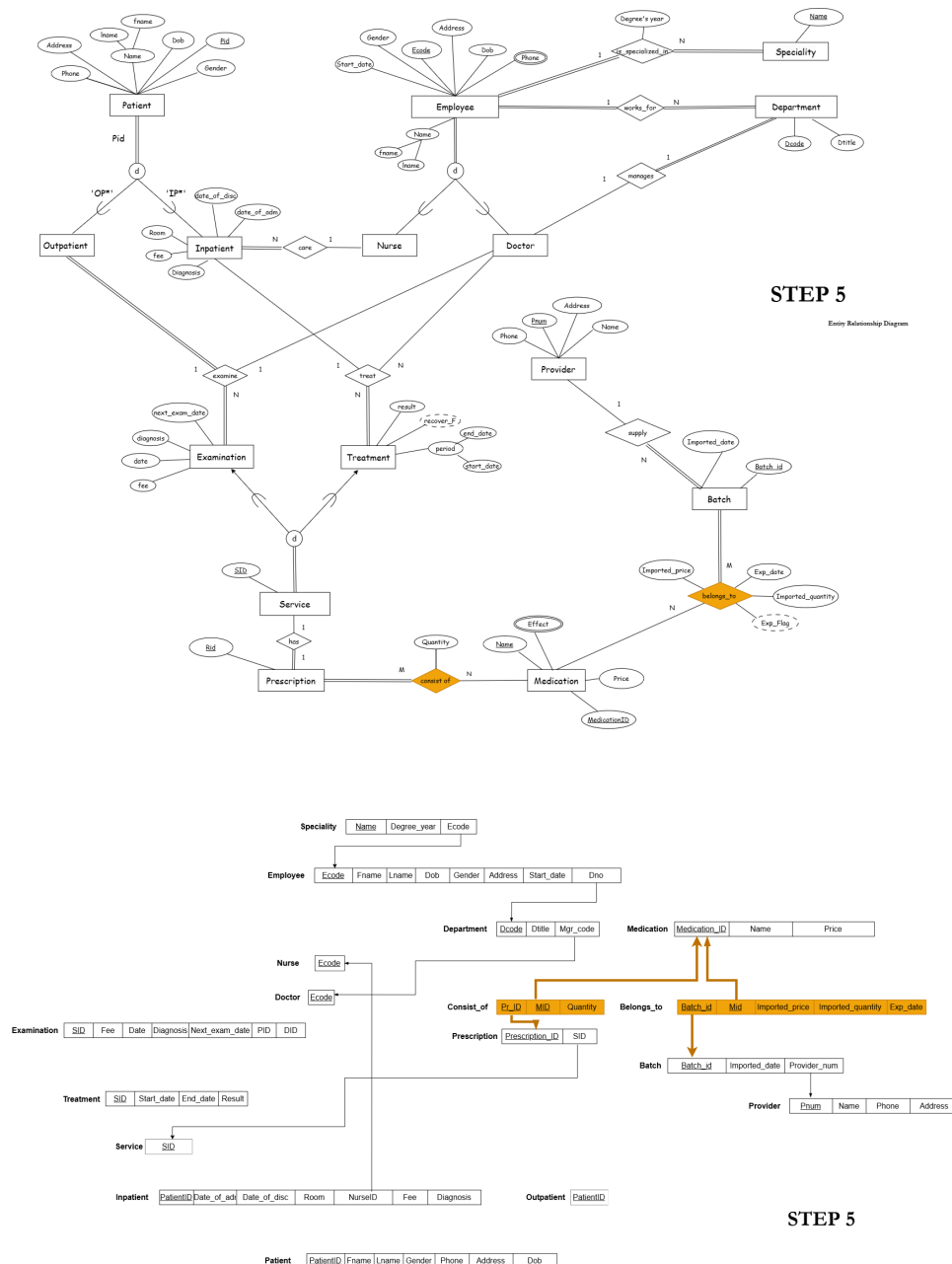


Figure 3.4: Mapping of Binary M:N Relationship Types

When an entity has multivalued attributes, we create a separate relation for each multivalued attribute.



3.7 Mapping of N-ary Relationship Types

For N-ary relationships, we map them to a new relation. The primary keys of all participating entities are included as foreign keys in this new relation.

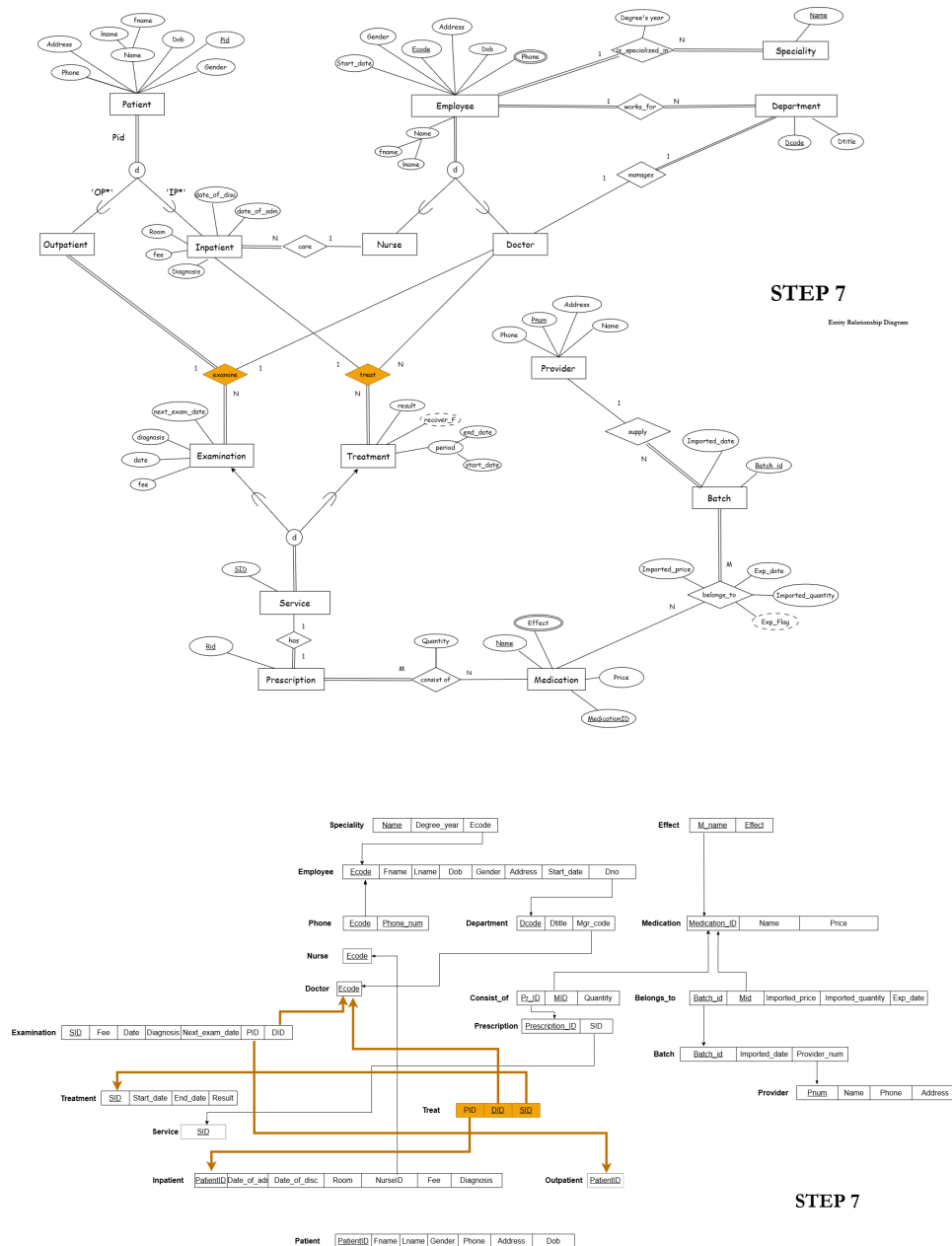


Figure 3.6: Mapping of N-ary Relationship Types

3.8 Mapping Specialization or Generalization

Specialization or generalization can be mapped using one of several strategies. Here we choose option 8A for all of our mapping.

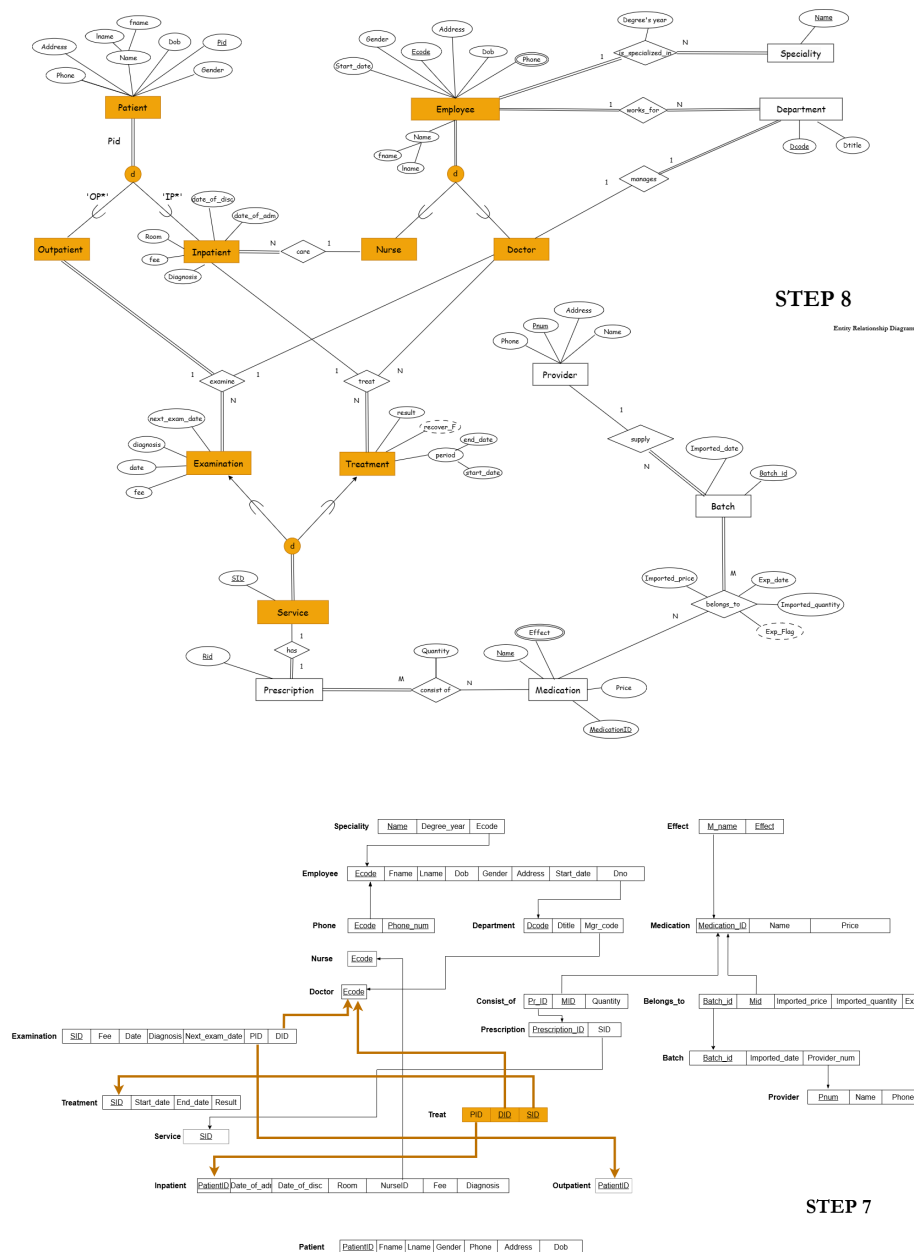


Figure 3.7: Mapping Specialization or Generalization

3.9 Mapping of Union Types (Categories)

There are none of Union Types in our EERD.



Section 4

Discussion

This final section reflects on the database design and discusses challenges, improvements, and future extensions that may enhance the system.

Managing Information and Workflow

Designing a hospital database involves managing large volumes of information, such as patient records, staff details, and medical supplies. Ensuring that entities and relationships are clearly defined is crucial for effective workflows.

Data Integrity

Maintaining data integrity is essential, especially when dealing with sensitive patient information. This database ensures that records are securely stored and easily accessible.

Scalability

Scalability is also important, as the hospital database must handle increasing data volumes without slowing down performance. The translation of the EER model into a relational schema ensures efficient querying and data storage.

Avoiding Complexity

The decision to avoid relationships with more than three entities prevents unnecessary complexity. The system remains easy to maintain and flexible.

Security and Privacy

Security and privacy are key concerns, with measures like role-based access control and encryption implemented to safeguard sensitive data.